

WebRTC Browser Calling — Fix & Test Summary

Real-world test follow-up · 28 September 2026 · browser-calling-5194.twil.io

Brett's 17-scenario real-world test pass surfaced five reliability issues to fix before a limited customer trial. All five have been addressed in [assets/index.html](#) and individually verified against the live Twilio deployment.

Fixes & Test Results

#	Issue	Fix	Result
1	Mobile calls locked to speakerphone, no way to switch	Output-device toggle. Desktop: full switching via setSinkId . Android: falls back to switching the <i>microphone</i> , which also flips the audio route (setSinkId isn't supported on mobile browsers).	Confirmed — desktop & Android (headset only) See limitation below re: earpiece.
2	Mic denial fails silently, no error shown	Pre-flight mic permission check; clear on-screen banner with platform-specific recovery steps (iOS/Android/desktop); Try Again button.	Confirmed working
3	Wifi → cellular handoff drops the call	Wired up the Voice SDK's reconnecting/reconnected events, previously unhandled, so the UI reflects the handoff instead of the call silently ending.	Confirmed working
4	Refreshing the browser mid-call drops the call	Leave-confirmation prompt on refresh/close; “tap to reconnect” banner shown on reload, tracked via sessionStorage tied to a genuine unload (pagehide) rather than any call-end event.	Confirmed working (two related bugs found & fixed during testing — see notes)
5	Desktop audio quality rated 3/5 vs WhatsApp/Meet	Set edge=sydney (routes via Twilio's NZ-local edge instead of default region selection) and explicit echo-cancellation / noise-suppression / auto-gain constraints.	Deferred — subjective, for Brett to re-judge by ear

Known Platform Limitation

“Switch to earpiece” cannot be delivered on mobile web at all. This was confirmed via a still-open Chromium issue (crbug 41276355): Chrome for Android exposes no API for a web page to route WebRTC audio to the earpiece — only speakerphone or an already-connected headset are reachable. iOS Safari exposes no audio-routing control to web pages whatsoever. The fix above is the practical ceiling for a browser-only tool; true earpiece support would require a native app wrapper, which was out of scope.

Deployment Status

All fixes are committed to **main** and live on **browser-calling-5194.twil.io**. Along the way, the GitHub Actions auto-deploy workflow was found to be non-functional (the workflow file was nested one directory too deep to be recognised by GitHub Actions) and has been relocated so future pushes to **main** deploy correctly.

Action needed: the customer-facing domain `call.wilderness.co.nz` is served by a separate, disconnected static copy in an S3 bucket behind CloudFront, last updated 14 May 2026. None of the fixes above reach real customers on that domain until this bucket's `index.html` is manually re-uploaded and the CloudFront cache invalidated.

Next Steps

- Re-upload **index.html** to the `call.wilderness.co.nz` S3 bucket and invalidate the CloudFront cache.
- Decide on a longer-term fix so that domain can't silently drift from Twilio again — point it directly at the Twilio service, or add an S3 sync step to the deploy process.
- Brett to re-run the desktop audio-quality comparison (Chrome & Safari) against WhatsApp/Meet.